

Design of discrete-time current controllers for induction motor drives based on an individual channel analysis approach

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Abstract—In this work a design method for discrete-time current controllers in induction motor drives is addressed. The method of individual channel analysis and design is used in order to represent the IM model as two single-input single-output (SISO) linear systems. Then, based on the obtained transfer functions, the controllers are designed in order to control the original multiple-input multiple-output system as if they were two SISO systems decoupled from each other. Therefore, the method allows to apply classical control techniques for the design of the controllers and simplify the analysis of the system. By following this approach, simple deadbeat controllers are proposed, analyzed and validated by simulation results.

Index Terms—individual channel, multi-variable analysis, induction motor, current control, electric drive

I. INTRODUCTION

During the last 40 years, the induction machines (IM) have acquired great interest from the academic and industrial community due to their low maintenance cost, robustness, high reliability and high torque density [1]. Based on advances in power electronics and digital systems, the use of closed loop control systems in IM has been constantly increasing in industrial applications. In particular, field-oriented control (FOC) is the most commonly used control scheme in these applications. FOC allows to obtain a fast dynamic torque response and presents an improved steady-state characteristic [2]. In order to obtain an high performance FOC, the use of accurate and robust inner stator current control loops is a determinant factor [3]. The objective of the current controllers is to obtain response signals exactly equal to the reference signals to prevent current errors from affecting the cross coupling between motor torque and flux [4].

Current controllers in FOC can be implemented in the stationary or in the synchronous reference frame. The standard approach is to implement PI controllers in the synchronous frame to separately control the direct and quadrature axis currents. The advantage of this strategy is that regulators operate on DC quantities, allowing to maintain zero steady-state error due to the action of the integrator function. However, recent works show that an essentially identical performance can be achieved by stationary frame linear controllers. In addition, this strategy has the advantage of not relying on synchronous transformations,

which reduces the potential errors induced by the frame-transformations, caused by measurement errors or bad rotor flux angle estimations. Moreover, this strategy requires less computational complexity [3], [5]. Hence, in this work the stationary reference frame is used for the controllers design.

For the current controller design there are many methods proposed in literature. Among the most common methods are the hysteresis control [6], proportional and integral control (PI) [7] and predictive control. In turn, predictive control can be classified into two groups: a predictive model [8], [9] and deadbeat [10], [11].

All previously mentioned controllers can be implemented to control single-input single output (SISO) systems without restrictions. However, due to the fact that IM are multiple input and multiple output (MIMO) nonlinear systems the design of the current controllers is generally complicated. Interactions between the input/output variables in a MIMO system make difficult the independent adjustment of each controller [12]. For this reason, in present work the individual channel analysis and design (ICAD) approach is addressed in order to obtain a individualization of the MIMO system into several SISO systems [13]. Once the individual channels are obtained, the control problem is reduced to the design of several controllers using classical control tools for the SISO systems. The ICAD allows to analyze the coupling of individual channels without simplifications, direct cancellations or loss on information found in other methods such as decoupling networks [14]. The ICAD approach has a broad field of application, from the design of gas turbine controllers, distillation columns and reactors [12] to the design of controllers for the maintenance of depth in submarines [15].

The ICAD approach is used in this work for the design of the inner loop discrete-time current controllers in an IM drive. In particular, the deadbeat control strategy is used [16]. With the proposed approach, two effective current controller were obtained that do not require manual or empiric gain adjustment. The performance of the designed controllers is demonstrated by simulation results.

The rest of the work is organized as follows. Section II describes the model of the IM in the stationary reference frame and addresses the ICAD approach. Then, in Section III the deadbeat design of the current controllers is shown. Finally,

Section IV shows the performance and coupling analysis of the current controllers as well as the FOC speed control loop performance.

II. ICAD APPROACH

The IM model in the stationary reference frame is described in this section. Then, the ICAD approach is followed to obtain the description of the IM as two SISO systems with coupling channels.

A. IM model

The IM is described in the stationary reference frame, denoted by (α, β) , as follows:

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}(\omega) \mathbf{x} + \mathbf{B} \mathbf{u}_{\alpha\beta} \\ \mathbf{y} &= \mathbf{C} \mathbf{x}\end{aligned}\quad (1)$$

where

$$\begin{aligned}\mathbf{A}(\omega) &= \begin{bmatrix} -a & 0 & bc & b\omega \\ 0 & -a & -b\omega & bc \\ L_M c & 0 & -c & -\omega \\ 0 & L_M c & \omega & -c \end{bmatrix}; \mathbf{B} = \begin{bmatrix} d & 0 \\ 0 & d \\ 0 & 0 \\ 0 & 0 \end{bmatrix}; \\ \mathbf{C} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}; \\ a &= \frac{L_r^2 R_s + L_m^2 R_r}{\sigma L_r}; b = \frac{L_m}{\sigma}; c = \frac{R_r}{L_r}; \\ d &= \frac{L_r}{\sigma}; e = L_m c; \sigma = (L_r L_s - L_m^2); \end{aligned}\quad (2)$$

Here, $\mathbf{x} = [i_\alpha, i_\beta, \lambda_\alpha, \lambda_\beta]$ represents stator currents and rotor flux linkages, respectively, $\mathbf{u}_{\alpha\beta} = [u_\alpha, u_\beta]$ represents the stator voltages, R_s, R_r, L_s, L_r represents the stator and rotor resistances and inductances, respectively, and L_m represents the mutual inductance. The mechanical system is described as:

$$\begin{aligned}\dot{\omega} &= \frac{P}{2J} (T_E - T_L) \\ T_E &= \frac{3P}{2} \frac{L_m}{L_r} [\lambda_\alpha i_\beta - \lambda_\beta i_\alpha]\end{aligned}\quad (3)$$

where, J represents the rotor inertia, P is the number of poles, ω represents the rotor angular speed, T_E and T_L are the electrical motor and load torque, respectively.

Thus, the IM is represented by an electrical (1) and a mechanical (3) subsystem. Through the electric subsystem, the stator voltages are controlled to control the stator currents and control the variables of the mechanical subsystem (T_E, ω). Because the dynamics of the mechanical subsystem vary slowly compared to the closed loop electrical subsystem, the speed could be considered as slowly varying for the inner current control loop design [2]. This is an accepted property in this application, which allows to design linear controllers for the inner control loops. Taking into account this consideration, current controllers for subsystem (1) are designed in this work, where $\mathbf{y}$ represents the outputs to be controlled.

B. ICAD applied to IM drives

As it is represented by (1), the IM can be considered as a two-input two-output system, with the (α, β) voltages representing the inputs and the (α, β) currents the outputs. The objective is to represent the IM as two individual SISO systems with the coupling represented as perturbations. As a difference from previous results, discrete-time controllers will be designed for each obtained SISO system [13].

With this approach, the electric model of the IM (1) is represented with transfer functions as

$$\begin{aligned}\begin{bmatrix} i_\alpha(s) \\ i_\beta(s) \end{bmatrix} &= \begin{bmatrix} g_{11}(s) & g_{12}(s) \\ g_{21}(s) & g_{22}(s) \end{bmatrix} \begin{bmatrix} u_\alpha(s) \\ u_\beta(s) \end{bmatrix} \\ \mathbf{G}(s) &= \begin{bmatrix} g_{11}(s) & g_{12}(s) \\ g_{21}(s) & g_{22}(s) \end{bmatrix}\end{aligned}\quad (4)$$

where $\mathbf{G}(s)$ is the set of transfer functions of the IM model, considering that the speed is constant to be able to represent the model as a linear system. This set of functions is obtained as

$$\mathbf{G}(s) = \mathbf{C} (s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B}\quad (5)$$

From (2) and (5) it can be obtained:

$$\begin{aligned}g_{11}(s) &= \frac{(n_{11}) s^3 + (n_{12}) s^2 + (n_{13}) s + (n_{14})}{s^4 + (d_{11}) s^3 + (d_{12}) s^2 + (d_{13}) s + (d_{14})} \\ g_{12}(s) &= -\frac{(n_{15}) s}{s^4 + (d_{11}) s^3 + (d_{12}) s^2 + (d_{13}) s + (d_{14})} \\ g_{21}(s) &= \frac{(n_{15}) s}{s^4 + (d_{11}) s^3 + (d_{12}) s^2 + (d_{13}) s + (d_{14})} \\ g_{22}(s) &= \frac{(n_{11}) s^3 + (n_{12}) s^2 + (n_{13}) s + (n_{14})}{s^4 + (d_{11}) s^3 + (d_{12}) s^2 + (d_{13}) s + (d_{14})}\end{aligned}\quad (6)$$

with

$$\begin{aligned}n_{11} &= d \\ n_{12} &= d(a + 2c) \\ n_{13} &= d(2ac + c^2 + \omega^2 - bce) \\ n_{14} &= d(ac^2 + a\omega^2 - bc^2e - be\omega^2) \\ n_{15} &= d(b\omega) \\ d_{11} &= (2a + 2c) \\ d_{12} &= (a^2 + 4ac + c^2 - 2bec + \omega^2) \\ d_{13} &= (2a^2c + 2ac^2 - 2beac + 2a\omega^2 - 2bec^2 - 2be\omega^2) \\ d_{14} &= (a^2c^2 + a^2\omega^2 - 2abc^2e - 2abe\omega^2 + b^2c^2e^2 \\ &\quad + b^2e^2\omega^2).\end{aligned}$$

In (6), $g_{11}(s)$, $g_{12}(s)$, $g_{21}(s)$ and $g_{22}(s)$ represent the individual transfer function, which depends on IM parameters and the rotor speed ω .

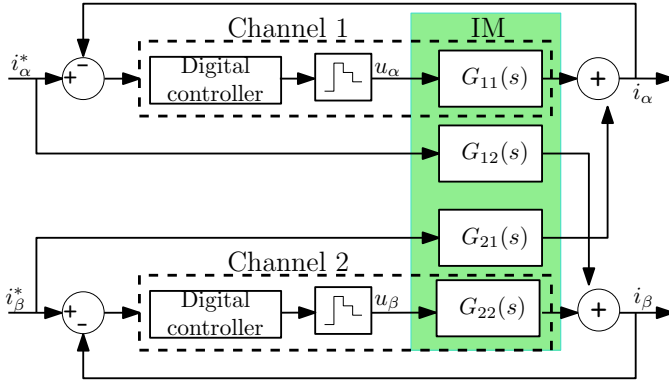


Fig. 1. Equivalent IM model diagram.

The individual transfer functions $g_{ij}(s)$ from (6), with $i, j = 1, 2$, are then used to obtain the equivalent model transfer functions $G_{ij}(s)$ as

$$\begin{aligned} G_{11}(s) &= g_{11}(s) (1 - \gamma(s) h_2(s)) \\ G_{12}(s) &= g_{12}(s) h_2(s) g_{22}^{-1}(s) \\ G_{21}(s) &= g_{21}(s) h_1(s) g_{11}^{-1}(s) \\ G_{22}(s) &= g_{22}(s) (1 - \gamma(s) h_1(s)) \end{aligned} \quad (7)$$

$$\gamma(s) = \frac{g_{ij}(s) g_{ij}(s)}{g_{ii}(s) g_{jj}(s)} ; \quad (8)$$

$$h_i(s) = \frac{k_i(s) g_{ii}(s)}{1 + k_i(s) g_{ii}(s)} ; \quad (9)$$

where $\gamma(s)$ represents the multivariable structure function of the original IM model, $h_i(s)$ represent the closed loop transfer functions of the plant $g_{ii}(s)$ obtained from the original IM system, k_i represent the controllers of each SISO system to be designed, $G_{11}(s)$ and $G_{22}(s)$ represent the decoupled transfer functions of the equivalent model, $G_{12}(s)$ and $G_{21}(s)$ represent the coupling transfer functions of the equivalent model.

The resulting equivalent IM model is illustrated in Fig. 1, where the sets obtained by the controller blocks and the transfer functions $G_{11}(s)$ and $G_{22}(s)$ represent the individual channels.

As a result of this representation, the obtained two individual channels $c_i(s)$ are described by

$$\begin{aligned} c_1(s) &= k_1(s) G_{11}(s) \\ c_2(s) &= k_2(s) G_{22}(s) \end{aligned} \quad (10)$$

where each input $u_\alpha(s)$ and $u_\beta(s)$ is related to one of the outputs $i_\alpha(s)$ and $i_\beta(s)$.

III. CONTROLLER DESIGN

The standard deadbeat digital control tools were selected for the design of the controller for each channel. One of the main characteristics of this control method is that the output of the controlled system reaches the desired reference with zero

error in a minimum finite time and without overshoot. The controller function is obtained based on the IM parameters, without additional gains adjustments, as follows:

$$k_i(z) = \frac{1}{G_{ii}(z)} \left[\frac{H(z)}{1 - H(z)} \right] \quad (11)$$

where $G_{ii}(z) = Z\{G_{ii}(s)\}$ represents the discrete open loop transfer function of the channel plant and $H(z)$ is a discrete function that is selected based on reference waveform that must be followed. In particular, a parabolic reference is used in this work [16].

It should be noted that in (11), the model transfer functions $G_{ii}(z)$ depend on the speed of the rotor ω and the IM parameters. For the proposed current control, the controllers functions are obtained based on the measured rotor speed. As an example, using the IM and control parameters listed in Table I, with rated speed, from (7) and by applying poles simplifications it can be obtained

$$\begin{aligned} G_{11}(s) &= \left(\frac{-0.05775s + 87.91}{s + 98.02} \right) \\ G_{12}(s) &= \left(\frac{0.1897s}{s + 4330} \right) \\ G_{21}(s) &= \left(\frac{-0.1897s}{s + 4330} \right) \\ G_{22}(s) &= \left(\frac{-0.05775s + 87.91}{s + 98.02} \right) \end{aligned} \quad (12)$$

Then, for this condition, the following controller transfer function is obtained

$$\begin{aligned} k_1(z) &= k_2(z) \\ k_2(z) &= 2.2018 \frac{(z^2 - 2z + 1)}{(z^2 - 1.999z + 1)}. \end{aligned} \quad (13)$$

It can be seen that the resulting controllers functions are equal. By following a root locus analysis, it can be deduced that the trajectory of the poles of the plant lies within the unit circumference which indicates the stability of the system [16]. From a practical point of view, this type of controller has the disadvantage of being sensitive to parametric variations and it can require high energy values to converge in a sample to the reference value, which can saturate the control. Therefore, the performance against parametric variations and perturbations introduced by the coupling channels for different rotor speeds is analyzed in the following section.

IV. CURRENT CONTROL PERFORMANCE OBTAINED BY USING THE ICAD APPROACH

Simulation results are shown in this section in order to analyze the designed current controllers performance. First, the inner current control loops are tested by using an arbitrary rotor speed in the IM model. Then, the performance of an outer speed control loop is analyzed when using a FOC strategy and arbitrary torque loads. IM and controls parameters are listed in

TABLE I
CONTROL AND IM PARAMETERS

Controller			
Sample frequency		10 kHz	
IM			
Power		5 Hp	
Frequency		60 Hz	
Line voltage		460 V	
Rated speed		1750 RPM	
R_s	1.115 Ω	R_r	1.083 Ω
L_{ls}, L_{lr}	5.974 mH	L_m	0.2037 mH
P	2	J	0.02 kgm ²

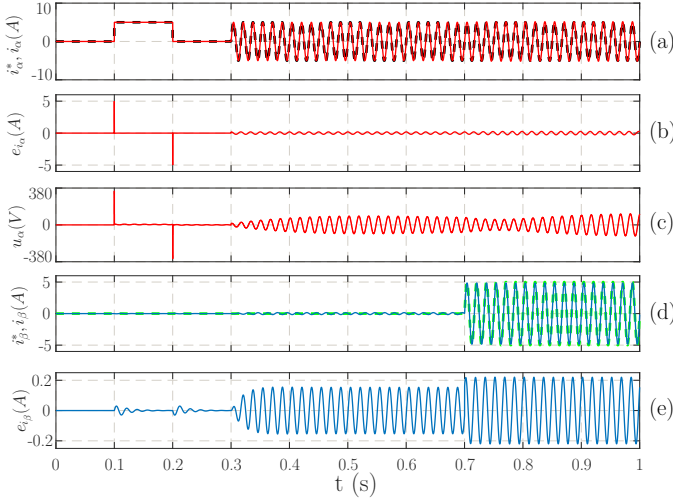


Fig. 2. Current control analysis. (a) Current i_α (b) i_α current tracking error (c) Voltage u_α (d) Current i_β (e) i_β current tracking error.

Table I. All simulations were performed using the MATLAB Simulink software.

A. Inner current control loop performance

Simulation results shown in Fig. 2 where obtained in order to evaluate the controllers performance using the model represented by (1) and the controllers from (13).

At rated rotor speed, a signal that begins with a step from 0 to 5.5 A at $t = 0.1$ s is applied in reference i_α^* . In $t = 0.2$ s, i_α^* changes from 5.5 A to 0 A. Then a sinusoidal signal with a 5.5 A peak value, with rated frequency, is applied at $t = 0.3$ s until $t = 1$ s. In turn, at $t = 0.7$ s a similar sinusoidal reference i_β^* is applied.

In Fig. 2 (a) and (d), dashed lines were used for the references and continuous lines for the response signals. The tracking error e_{i_α} of current i_α is shown in Fig. 2 (b), where it can be seen that a transient is produced during the sinusoidal part of i_α^* that grows until stabilizing in a peak value of 0.2 A. Fig. 2 (e) shows the tracking error e_{i_β} of current i_β which presents small oscillations during the variations of the signal i_α^* , due to the coupling effect, with 0.18 A peak values in steady state. At $t = 0.7$ s this tracking error increases up to 0.2 A after the introduction of the sine reference in i_β^* .

Fig. 2 (c) shows that voltage u_α from the controller saturates in 380 V during the transients of the step reference in i_α^* and for the sinusoidal reference from $t = 0.3$ s grows until it stabilizes with a peak value of 160 V. Then, at $t = 0.7$ s a transient is produced due to the cross coupling effect.

From simulation results shown in Fig. 2, it can be seen that the controllers have a good performance although they are affected by the disturbances produced due to the coupling effect. The maximum obtained tracking errors are considered small enough for the present work.

B. Coupling analysis

The coupling levels between channels $c_1(s)$ and $c_2(s)$ can be evaluated with the following relations, called sensitivity functions:

$$\begin{aligned} \frac{i_\alpha}{i_\beta^*(s)} &= \frac{1}{1 + c_1(s)} \frac{g_{12}(s)}{g_{22}(s)} h_2(s) \\ \frac{i_\beta}{i_\alpha^*(s)} &= \frac{1}{1 + c_2(s)} \frac{g_{21}(s)}{g_{11}(s)} h_1(s). \end{aligned} \quad (14)$$

Through the analysis by frequency response of the sensitivity functions (14) it is possible to analyze the MIMO control response as if they were two SISO systems. Fig. 3 shows the Bode diagram of the cross coupling of input 1 with respect to output 2 (which is equivalent to the cross coupling input of 2 with respect to output 1 due to the symmetry of the system) based on the values expressed in Table I for different rotor speeds. It can be seen that as ω increases, the coupling level also increases. Analytically, the increase in coupling implies measurement noise in channel 2 which increases the gain over channel 1 [17]. In this way, it can be seen that in the worst case the magnitude of the coupling is limited to a maximum level of -30 dB. As it was reported, it is considered that with these levels the individual channels are effectively decoupled without the use of additional decoupling methods [14]. In addition, it can be observed that the peaks of sensitivity are produced at frequencies near the rotor angular speed ω , which shows that an oscillatory behavior can be expected at high frequencies [18].

In general terms, when designing the individual controllers for each channel, the analysis is reduced to the design of the $k_i(s)$ controllers for the $G_{ii}(s)$ plants. Once the design has been made, it is possible to verify the effects of the coupling which acts as a disturbance.

C. Speed control loop performance using designed controllers

Simulations of a standard FOC were performed to analyze the operation of the proposed current controller in a speed control loop [8]. It should be mentioned that the design of the speed control loop is not studied in depth, since the present work focuses on the design of the current controllers. The simulations were performed using the control scheme illustrated in Fig. 4.

Obtained results are shown in Fig. 5. The IM starts from the idle state with a load torque of 4 Nm and a speed step reference of 1750 RPM, as shown in Fig. 5 (a). The speed

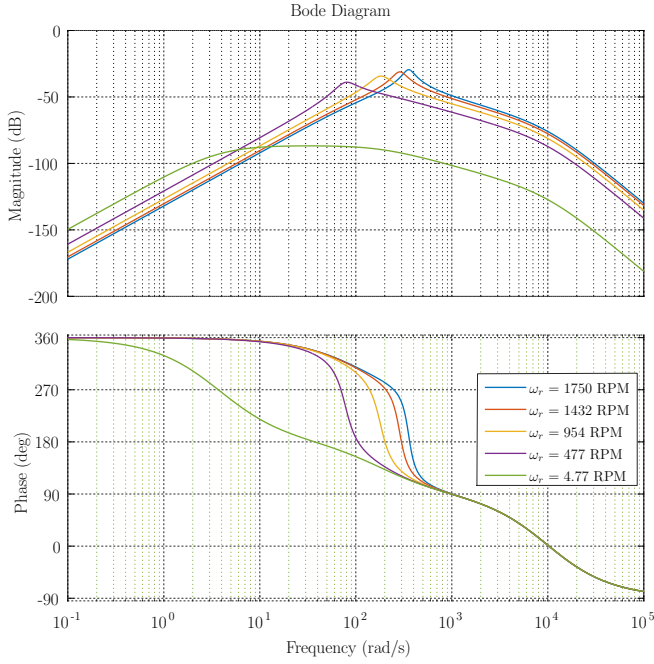


Fig. 3. Bode diagram of coupling sensitivity functions.

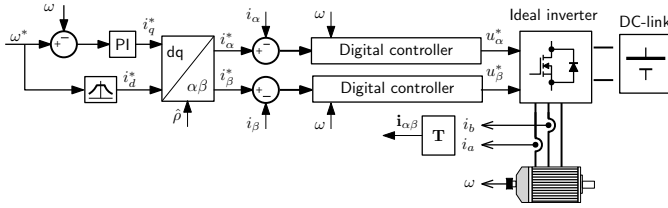


Fig. 4. FOC speed control scheme.

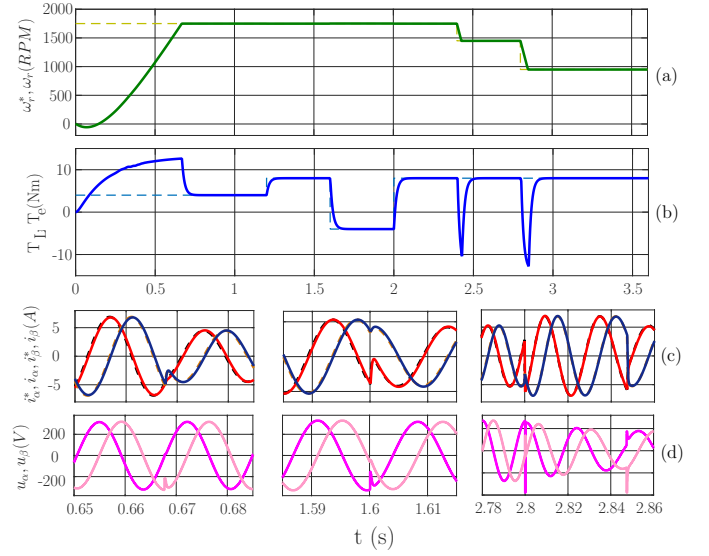
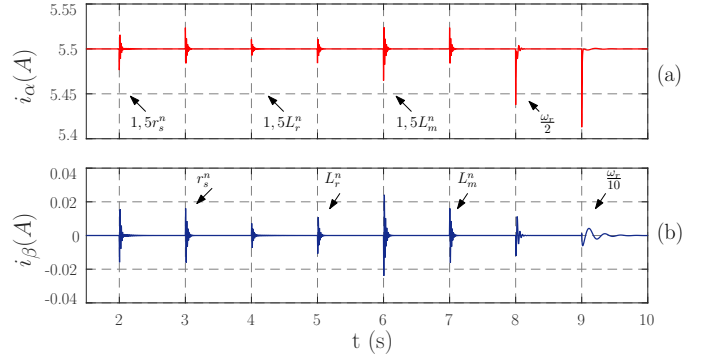
reference is reached at $t = 0.65$ s, where the electrical torque equals the load torque, as shown in Fig. 5 (b).

In $t = 1.2$ s the load torque is increased to 8 Nm, then in $t = 1.6$ s is decreased to -4 Nm and by $t = 2$ s is returned to 8 Nm. During these load torque variations, the speed reference remains constant. At $t = 2.4$ s the speed reference is decreased by 300 RPM and at $t = 2.8$ s it is again decreased by 450 RPM. After the last speed reference variation, the control reaches the target value with a response time of 0.1 s.

The detail of the stator currents and voltages, respectively, of some intervals of the test are shown in Fig. 5 (c) and (d).

In Fig. 5 (c) it can be seen that the current controllers follow the references efficiently. In addition, Fig. 5 (d) shows that voltages u_α and u_β remains within the working limits, except at $t = 2.8$ s and $t = 2.848$ s where saturations are produced, but instantly voltages return to the normal operating condition.

Based on these simulation results, it can be concluded that the FOC speed control presents suitable characteristics for speed tracking and against load torque variations when using the proposed current controllers in the inner control loop.


 Fig. 5. FOC speed control performance using the designed deadbeat controllers. (a) Rotor speed ω (b) Motor and load torque, T_E and T_L (c) Currents i_α and i_β (d) Voltages u_α and u_β .

 Fig. 6. Parameter variations test results. (a) Current i_α (b) Current i_β .

D. Performance of the current controllers against parameters variations

The simple standard designed deadbeat controllers have the disadvantage of being strongly dependent on the model of the plant and therefore sensitive to parametric variations. Therefore, a parametric variations study was performed and the results presented in Fig. 6 were obtained. In this study, the current controllers were calculated for rated speed and frequency and sudden variation were applied in IM parameters as well as variations of the rotor speed using the control scheme from Fig. 1.

An i_α^* sine waveform was applied to the control, with a nominal frequency and peak values of 5.5 A. In addition, a constant $i_\beta^* = 0$ A was applied to show the coupling effect in channel i_β^* when variations occur in channel i_α^* . In $t = 2$ s a variation of the stator resistance of $r_s = 1.5r_s^n$ is applied, where the superscript n refers to the rated values. At $t = 3$ s the stator resistance returns to the rated value $r_s = r_s^n$. The stator resistance produces a peak tracking error of 0.02 A

followed by decreasing undulations in the currents. Similarly, at $t = 4$ s a variation of the rotor inductance $L_r = 1.5L_r^n$ is applied during 1, s, and the is returned to the rated value. This variation produces a peak of 0.01 A followed by a decreasing ripple in the current tracking errors. During $6\text{ s} \leq t \leq 7\text{ s}$ the variation of the mutual inductance $L_m = 1.5L_m^n$ is applied, which produces a transient with a peak of 0.035 A in the current tracking errors. A rotor speed variation $\omega = 0.5\omega^n$ is applied at $t = 8$ s, and a variation $\omega = 0.1\omega^n$ at $t = 9$ s. The speed variations produces a maximum current tracking error of 0.08 A.

Presented results shows that the designed deadbeat controllers introduce transients in the current tracking errors during sudden parameters and measured speed errors. However, these errors converge to a null value in a short time.

V. CONCLUSIONS

In this work, a current control scheme was presented using an individual channels analysis and design approach. It was shown that standard discrete-time deadbeat controllers can be designed based on obtained individual channels representations of the induction motor model in the stationary reference frame. From the obtained scheme, it was demonstrated that two symmetrical controllers are required to control the stator currents as two SISO systems. Although a reduced decoupling sensitivity levels were obtained, the resulting coupling effect is perceptible in the obtained simulations results. Nevertheless, the addressed design method presents the advantage that the controller is simple to implement, its parameters can be obtained from the IM parameters and no manual gain adjustment is required. A good performance was obtained when incorporating the current controls in a standard field oriented speed control loop during simulation results. In addition, the designed current controllers showed a good performance against parameter variations and speed measurement errors, with a null steady state tracking error. In future work, comparisons of different control strategies designed following the proposed methodology will be carried out.

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